

B-9-X

Roll No.

Total No. of Questions : 4]

[Total No. of Printed Pages : 8

12thARF(SZ)JKUT2024-25**309-X****CHEMISTRY****Time : 3 Hours]****[Maximum Marks : 70****General Instructions :**

- (i) There are total four Sections in the question paper. All questions are compulsory.
- (ii) **Section-A** contains 10 Objective Type Questions (Multiple Choice Questions) of 1 mark each. $1 \times 10 = 10$ marks
- (iii) **Section-B** contains 9 Very Short Answer Type Questions of 2 marks each to be answered in 20-30 words.
 $2 \times 9 = 18$ marks
- (iv) **Section-C** contains 9 Short Answer Type Questions of 3 marks each to be answered in 100-150 words. $3 \times 9 = 27$ marks
- (v) **Section-D** contains 3 Long Answer Type Questions of 5 marks each to be answered in 150-200 words. $5 \times 3 = 15$ marks
- (vi) Use log table if necessary. Use of scientific calculators is not allowed.

12thARF(SZ)JKUT2024-25-309-X

Turn Over

B-9-X

(2)

SECTION-A

1 each

**OBJECTIVE TYPE QUESTIONS
(MULTIPLE CHOICE QUESTIONS)****1. Select the correct one :**

(i) The molal elevation constant is the ratio of the elevation in B.P. to :

(A) Molarity

(B) Molality

☒ (C) Mole fraction of solute

(D) Mole fraction of solvent

(ii) Which of the following solutions will have highest boiling point at 1 atm pressure ?

(A) 0.1 m FeCl_3

☒ (B) 0.1 m BaCl_2

(C) 0.1 m NaCl

(D) 0.1 m Urea (NH_2CONH_2)

(iii) Nernst equations give effect of :

- (A) Temperature on EMF
- (B) Concentration on EMF
- (C) Pressure of the gas (if any) on EMF

☒ (D) All the three above

(iv) The rate constant of a reaction is $2.5 \times 10^{-2} \text{ sec}^{-1}$. The order of the reaction is :

☒ (A) One

(B) Zero

(C) Two

(D) Three

(v) S_N^1 reactions occur through the intermediate formations of :

☒ (A) Carbocation

(B) Carbanions

(C) Free radical

(D) None of these

(vi) On heating with Sodalime salicylic acid gives :

- ☒ (A) Phenol
- (B) Benzoic acid
- (C) Sodium salicylate
- (D) Benzene

(vii) Reduction of nitroethane with LiAlH_4 gives :

- (A) Ethyl hydroxylamine
- (B) Ethanol
- ☒ (C) Ethylamine
- (D) Ethanamide

(viii) A complex in which dsp^2 hybridisation takes place is :

- ☒ (A) Square planar
- (B) Tetrahedral
- (C) Triangular planar
- (D) Pyramidal

(5)

(ix) Adenosine is example of a :

(A) Nucleotide

~~(B) Nucleoside~~

(C) Purine base

(D) Pyridine base

(x) Which of the following biomolecules contains a non-transition metal ion ?

(A) Haemoglobin

~~(B) Chlorophyll~~

(C) Insulin

(D) Vitamin B₁₂

SECTION-B

2 each

VERY SHORT ANSWER TYPE QUESTIONS

2. (i) Calculate the half-life of a first order reaction from their rate constants given below :

(a) 200 s^{-1}

(b) 2 min^{-1}

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B-9-X

Turn Over

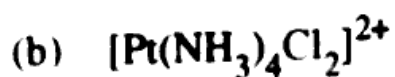
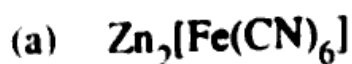
(iii) Why does a transition series consist of 10 elements ?

(iii) Why is an alkylamine more basic than ammonia ?

(iv) What is Lucas reagent ?

(v) What are essential and non-essential amino acids ?

(vi) Write the IUPAC names of :



(vii) Define rate of reaction and rate constant.

(viii) Why cannot molecularity of any reaction be equal to zero ?

(ix) Give a brief account of Wurtz reaction.

SECTION-C

3 each

SHORT ANSWER TYPE QUESTIONS

3. (i) What do you mean by a Nucleophilic addition reaction ? Give its mechanism.

(ii) Define Kohlrausch's law. How can it be used to find the degree of dissociation of a weak electrolyte ?

(7)

✓(iii) Explain, why transition metal ions usually show paramagnetic behaviour ?

(iv) What happens when acidified KMnO_4 reacts with (a) H_2S
(b) Oxalic acid (c) KNO_2 ?

✓(v) Define cis and trans isomerism. Draw the cis and trans isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ ion. <https://www.jkboseonline.com>

(vi) How will you convert ethyl bromide to :

(a) Ethoxyethane

(b) Ethane

(c) Ethanol

✓(vii) How will you distinguish between primary, secondary and tertiary alcohols by Victor Meyer's test ? Give chemical reactions.

✓(viii) What is Diazotisation ? Discuss its mechanism.

✓(ix) What are Carbohydrates ? How are they classified ?

(8)

SECTION-D

5 each

LONG ANSWER TYPE QUESTIONS

- (i) State and explain Raoult's law for (a) Volatile solute (b) non-volatile solute.

Or

What do you understand by Colligative properties of a solution ?
Explain briefly osmosis and osmotic pressure.

- (ii) What is an electrochemical cell ? Describe working of Daniell cell.

Or

What is Corrosion ? Give mechanism (electrochemical phenomenon) of rusting of iron. What do you understand by cathodic and barrier protection of corrosion ?

- (iii) Describe any five general methods of preparations of carboxylic acids.

Or

Giving chemical equations write a brief account of the following :

- (a) Aldol condensation
- (b) Cannizzaro reaction
- (c) Rosenmund reduction

B-9-X

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12thARM(SZ)JKUT2024**1109-X****CHEMISTRY****Time : 3 Hours]****[Maximum Marks : 70****General Instructions :**

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12thARM(SZ)JKUT2024-1109-X**B-9-X**

Turn Over

(2)

SECTION-A

1 each

**OBJECTIVE TYPE QUESTIONS
(MULTIPLE CHOICE QUESTIONS)**

1. Select the correct one :

(i) In which mode of expression the concentration of solution remain independent of temperature ?

(A) Molarity

(B) Normality

(C) Formality

☒ (D) Molality

(ii) If an aqueous solution of glucose is allowed to freeze, then crystals of which will be separated out first ?

(A) Glucose

☒ (B) Water

(C) Both of these

(D) None of these

(iii) The amount of an ion liberated on an electrode during electrolysis does not depend upon :

(A) Current strength

(B) Conductance of the solution

(C) Time

☒ (D) Electrochemical equivalent of the element

(3)

(iv) Collision theory is applicable to :

- (A) First order reaction
- (B) Zero order reaction
- ☒ (C) Bimolecular reaction
- (D) Intramolecular reaction

(v) Alkyl halides undergoing nucleophilic bimolecular substitution involve :

- (A) Formation of carbocation
- (B) Racemic mixture
- ☒ (C) Inversion of configuration
- (D) Retention of configuration

(vi) Among the following compounds strongest acid is :

- ☒ (A) $\text{HC} \equiv \text{CH}$
- (B) C_6H_6
- (C) C_2H_6
- (D) CH_3OH

(vii) The weakest base among the following is :

- ☒ (A) Dimethylamine
- (B) Aniline ✗
- (C) Methylamine
- (D) Ethylamine

(4)

(viii) A transition metal exists in its highest oxidation state. It is expected to behave as :

- (A) A chelating agent
- (B) A central metal in a coordination compound
- ☒ (C) An oxidising agent
- (D) A reducing agent

(ix) The human body does not produce :

- (A) Enzymes
- (B) DNA
- ☒ (C) Vitamins
- (D) Hormones

(x) Adenosine is an example of :

- ☒ (A) Nucleotide
- (B) Nucleoside
- (C) Purine base
- (D) Pyrimidine base

SECTION-B

2 each

VERY SHORT ANSWER TYPE QUESTIONS

2. (i) What is the difference between Rate Law and Law of Mass Action ?
- (ii) What is meant by didentate and ambidentate ligands ?
- (iii) Why are alcohols less acidic than water ?
- (iv) What is diazotisation ?
- (v) Write *two* main functions of carbohydrates in plants.
- (vi) Write IUPAC names of :
- (a) $[\text{CrCl}_2(\text{en})(\text{NH}_3)_2]^+$
- (b) $\text{K}_3[\text{Fe}(\text{CN})_6]$
- (vii) Why molecularity is applicable only for elementary reactions and order is applicable for elementary and as well as complex reactions ?
- (viii) How does average rate of reaction differ from instantaneous reaction rate ?
- (ix) Why are haloarenes less reactive than haloalkanes towards nucleophilic substitution reactions ?

(6)

SECTION-C

3 each

SHORT ANSWER TYPE QUESTIONS

3. (i) Formic acid (methanoic acid) is stronger acid than acetic acid (ethanoic acid). Explain.
- (ii) Define conductivity and molar conductivity for the solution of an electrolyte.
- (iii) Explain the following about transition metals :
- (a) Magnetic behaviour
 - (b) Oxidation states
- (iv) How is potassium dichromate prepared from chromite ore ?
Give its three oxidising properties. <https://www.jkboseonline.com>
- (v) Discuss briefly giving an example in each case the role of co-ordination compounds in :
- (a) Biological system
 - (b) Medicinal chemistry
- (vi) How will you convert ethyl bromide to :
- (a) Ethane
 - (b) Ethoxyethane
 - (c) Ethanenitrile ?

(7)

- (vii) What are phenols ? How do they differ structurally from aromatic alcohols ?
- (viii) What is Hinsberg's reagent ? How will you distinguish between primary, secondary and tertiary amines by it ?
- (ix) What are α -amino acids ? How are they related to proteins ?
Give the structure of two amino acids ?

SECTION-D

5 each

LONG ANSWER TYPE QUESTIONS

4. (i) Define :
- (a) Mole fraction
 - (b) Molality
 - (c) Molarity

Calculate the mole fraction of ethylene glycol ($C_2H_6O_2$) in a solution containing 20% of $C_2H_6O_2$ by mass.

Or

Define and explain elevation in boiling point. How can you calculate the molecular mass of a non-volatile solute with it ?

(8)

(ii) Define Kohlrausch's law. How does it help in :

- (a) Calculation of λ° for a weak electrolyte
- (b) Degree of dissociation of a weak electrolyte ?

Or

What are fuel cells ? Describe $H_2 - O_2$ fuel cell.

(iii) Describe the following :

- (a) Esterification
- (b) Cannizzaro reaction
- (c) Cross aldol condensation
- (d) Decarboxylation

Or

- (a) Write *five* methods for the preparation of aldehydes.
- (b) How are aldehydes distinguished from ketones using Tollen and Fehling's reagents ? Give chemical reactions.

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12thARM(SZ)JKUT2024—1109—X

B-9-X

E-2-A

Roll No.

Total No. of Questions : 28]

[Total No. of Printed Pages : 7

XIIAPBIAJKLK23**9802-A****CHEMISTRY**

Time : 3 Hours]

[Maximum Marks : 70

Section-A

1 each

(Very Short Answer Type Questions)

1. What is body centred cubic ?
2. The zig-zag motion of colloidal particles is called
3. $\text{CH}_3\text{--CH}_2\text{--OH} \xrightarrow{\text{Cu/573K}} \text{CH}_3\text{CHO} + \text{H}_2$

4. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{--C--NH}_2 \\ | \\ \text{CH}_3 \end{array}$ is a primary amine. (True/False)

5. Write IUPAC name of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$.

XIIAPBIAJKLK23-9802-A

Turn Over

E-2-A

(2)

6. The disaccharide present in milk is :

(A) Sucrose

(B) Maltose

✓ (C) Lactose

(D) Cellobiose

7. Silver mirror is a test for :

(A) Aldehydes

(B) Thioalcohols

(C) Acids

(D) Ethers

Section-B

2 each

(Short Answer Type Questions-I)

8. What is Salt Bridge ? What is it used for ?

9. Calculate half life of a first\order reaction where the specific rate constant is :

(a) 200 s^{-1}

(b) 2 min^{-1}

XIIAPBIAJKLK23—9802-A

E-2-A

(3)

10. What are Transition Metals ? Which *d*-block elements are not considered as transition metals ? Why ?

Or

Transition metals have high melting and boiling points. Explain.

11. Define with one example the following :

- (A) Complex ion
- (B) Ligand
- (C) Coordination number
- (D) Oxidation State

Or

Define cis and trans isomers. Draw the cis and trans isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ ion.

12. Explain as to why haloarenes are much less reactive than haloalkanes towards nucleophilic substitution reactions ?
13. Write short note along with reaction on Friedel-Crafts reaction.

(4)

Section-C

3 each

(Short Answer Type Questions-II)

14. What are Stoichiometric defects in solids ? Explain Frenkel defect in ionic solids. What are the consequences of this defect in solids ?
15. Draw a neat and well labelled diagram of a dry cell. Is it really dry ?
Give reactions taking place at anode and cathode.
16. Briefly explain the effect of temperature on the rate constant of a reaction. <https://www.jkboseonline.com>
17. Define Gold number. How is it related to protective power of a colloid ?
18. Give a brief account of calcination and roasting.
19. Explain why acidic strength of oxo acids of halogens decreases while increasing the atomic number of elements.
20. What is lanthanide contraction ? What are its causes and consequences ?

XIIAPBIAJKLK23-9802-A

E-2-A

(5)

21. What happens when : (Give reactions only)

- (i) Phenol is heated with excess of concentrated nitric acid in presence of concentrated H_2SO_4
- (ii) Phenol reacts with bromine in aqueous medium

22. Illustrate the following reactions giving suitable examples in each case :

- (i) Diazotisation reactions
- (ii) Coupling reactions

Or

How is aniline prepared from nitrobenzene ? How will you convert aniline into (i) Benzonitrile (ii) Acetanilide ?

23. State the significance of primary and secondary structures of proteins.

Or

What are Vitamins ? Give three examples which vitamins are fat soluble and which are water soluble ? What is the importance of vitamin A and D ?

(6)

24. What are Polymers ? How are they classified on the basis of structure ?

Or

How is Bakelite prepared ? What are its major uses ?

25. What are Antibiotics ? Name *two* antibiotics which are specific for certain diseases.

Or

Explain Cleansing action of soap.

Section-D

5 each

(Long Answer Type Questions)

26. Illustrate with the help of diagrams different types of non-ideal solutions. Explain the reasons for negative and positive deviations.

Or

What do you mean by abnormal molecular mass ? Explain the factors with suitable examples which bring about the abnormality.

XIIAPBIAJKLK23-9802-A

E-2-A

(7)

27. Discuss the process of manufacture of ammonia by Habers process.

Draw the flow chart diagram for the process.

Or

Briefly describe the various allotropic forms of phosphorous. Also give method of preparation of phosphine.

28. How is acetaldehyde prepared from (i) Ethanol (ii) Acetic acid (iii) Acetylene ? Give reactions of acetaldehyde with :

(a) NaHSO_3

(b) HCN

(c) CH_3MgBr

Or

Describe the following reactions :

(a) Aldol condensation

(b) Cannizzaro reaction

(c) Rosenmunds reaction

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E-2-A

A-2-A

Roll No.....

Total No. of Questions : 28]

[Total No. of Printed Pages : 7

12thSZARJD22**6002-A****CHEMISTRY****Time : 2.30 Hours]****[Maximum Marks : 70****Section-A**

1 each

(Very Short Answer Type Questions)

1. What happens when a negatively charged colloidal sol is added to a positively charged colloidal sol ?
2. The IUPAC name of $[\text{Co}(\text{en})_3\text{Cl}_3]$ is
3. $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \xrightarrow{\text{No peroxide}} \text{CH}_3-\text{CH}_2-\text{CH}_2\text{Br}$,
is the major product. (True/False)
4. Arrange HCl, HBr and HI in the decreasing order of reactivity with alcohols.
5. Write IUPAC name of $\text{OHC}-\text{CH}_2-\underset{\text{CHO}}{\text{CH}}-\text{CHO}$.

12thSZARJD22-6002-A

Turn Over

A-2-A

(2)

6. Choose the correct answer :

Benzamide on treatment with POCl_3 gives :

- (A) Aniline
- (B) Benzonitrile
- (C) Chlorobenzene
- (D) Benzylamine

7. Give an example of Anomers.

Section-B

2 each

(Short Answer Type Questions-I)

- 8. What do you mean by Frenkel defect ?
- 9. What is Piezoelectricity ? Give one example of such solid.
- 10. What is equivalent conductivity ? Give its units.

Or

What is Secondary Cell ? Give one example.

- 11. What is meant by the terms average and instantaneous rates of reaction ? How are they expressed ?

Or

Define Rate Law.

12th SZARJD22-6002-A

A-2-A

(3)

12. Transition metals have ability to form alloys. Give reason.
13. Define Ionisation Isomerism in coordination compounds. Give *one* example.

Section-C

3 each

(Short Answer Type Questions-II)

14. Define Corrosion. Explain electrical protection for protecting iron from corrosion.
15. Define First Order Reaction. Derive rate equation for first order reaction. <https://www.jkboseonline.com>
16. What is Brownian Movement ? What is its cause and significance ?
17. Give a brief account of calcination and roasting.
18. Why does nitrogen differ from rest of the members of group 15 ?
Give *three* points of difference.
19. Explain, why haloarenes are much less reactive than haloalkanes towards nucleophilic substitution ?

(4)

20. Define and classify Vitamins. What is the role of Vitamin C in our daily life ? Give its sources.
21. What is Lanthanide contraction ? Give its causes.
22. How is phenol obtained from chlorobenzene and benzene diazonium chloride ? How phenol reacts with :
- (i) $\text{CHCl}_3 + \text{NaOH}$ at 340K
 - (ii) Conc. H_2SO_4 at 375K ?

Or

How can you prepare ethyl alcohol from :

- (a) Ethene
- (b) Ethyl bromide ?

How can you convert ethyl alcohol to :

- (i) Acetaldehyde
- (ii) Diethyl ether ?

23. Why are amines basic ? Compare basic strength of ammonia, ethylamine and diethylamine.

12th SZARJD22-6002-A

A-2-A

(5)

Or

How will you prepare ethylamine by using Hofmann bromide reaction and how will you convert ethylamine to ethylisocyanide ?

24. Give preparation and uses of Nylon-66 and Terylene.

Or

Explain the term co-polymerisation and give its *two* examples.

25. What are Tranquilizers ? Give *two* examples.

Or

What are Antibiotics ? Give *two* examples.

Section-D**5 each****(Long Answer Type Questions)**

26. Define the term Osmotic pressure. Describe how molecular mass of a substance can be determined on the basis of osmotic pressure. 200 cm³ of an aqueous solution of a protein contains 1.26 g of the protein. The osmotic pressure of such solution at 300K is found to be 2.57×10^{-3} bar. Calculate molecular mass of the protein.

(6)

Or

Define the following :

- (i) Molarity
- (ii) Molality
- (iii) Mole fraction

Concentrated nitric acid in laboratory work is 68% nitric acid by mass in aqueous solution. What should be the molarity of such a sample of the acid if the density of the solution is 1.504 gmL^{-1} ?

27. What are Interhalogen Compounds ? How are they classified ? Give the structure and mention the hybridisation of central atom of the following :

- (a) IF_7
- (b) BrF_3
- (c) BrF_5

Or

Describe the manufacture of sulphuric acid by contact process.

12th SZARJD22—6002—A

A-2-A

(7)

28. Describe the following reactions :

- (a) Aldol condensation
- (b) Cannizzaro reaction
- (c) Rosunmund's reaction

Or

How is acetic acid prepared from :

- (i) Acetylene
- (ii) Ethyl alcohol
- (iii) Acetamide ?

How does acetic acid react with :

- (i) Cl_2 in presence of phosphorus
- (ii) Methanol in presence of H_2SO_4 ?

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B-2-B

Roll No.....

Total No. of Questions : 28]

[Total No. of Printed Pages : 7

XIIKDAR21**5002-B****CHEMISTRY**

Time : 3 Hours]

[Maximum Marks : 70

Section-A**(Very Short Answer Type Questions)**

1 each

1. If the radius of octahedral void is ' r ' and the radius of the atoms in the close packing is ' R ', what is the relation between ' r ' and ' R ' ?
2. Identify the positively charged sol :
(A) Haemoglobin
(B) As_2S_3
(C) Clay
(D) Gold sol
3. Calculate the oxidation number of Co in $[CoBr_2(CN)_2]^+$.

XIIKDAR21-5002-B

B-2-B

Turn Over

(2)

4. Give the IUPAC name of $(\text{CH}_3)_3\text{C.OCH}_3$.
5. When a mixture of calcium acetate and calcium formate is distilled, the product formed is :
- (A) Formaldehyde
- (B) Acetone
- (C) Acetaldehyde
- (D) None of these
6. Predict the product of the following reaction :
- $$\text{C}_2\text{H}_5\text{NO}_2 \xrightarrow{\text{Sn/HCl}} \dots\dots\dots$$
7. Out of sucrose and maltose, sucrose is a reducing sugar.

(True/False)

Section-B**(Short Answer Type Questions-I)**

2 each

8. Define equivalent conductivity and give its S.I. units.
9. Define Activation Energy. How is it related to threshold energy?
10. Give any *two* oxidising reactions of KMnO_4 in acidic medium.

XIIKDAR21—5002-B

B-2-B

(3)

Or

Why do transition metals form interstitial compounds ?

11. Define the terms with examples :

(a) Coordination complex

(b) Coordination number

Or

Explain hydrate isomerism in coordination compounds with an example.

12. Give any two chemical reactions of chlorobenzene.

13. What are the important uses and environmental effects of D.D.T. ?

Section-C

(Short Answer Type Questions-II)

3 each

14. Define the following properties of solids :

(a) Paramagnetism

(b) Diamagnetism

(c) Ferromagnetism

XIIKDAR21-5002-B

B-2-B

Turn O

(4)

15. State and explain Faraday's second law of electrolysis.
16. Discuss the effect of temperature on the rate of reaction.
17. What is Chemical Adsorption ? Discuss the main characteristics of chemical adsorption.
18. Give a brief account of calcination and roasting.
19. Name the elements of group 15 and discuss their oxidation states.
20. What is Lanthanoid Contraction ? Explain the cause of lanthanoid contraction. <https://www.jkboseonline.com>
21. Give the chemical reactions of ethyl alcohol with :
 - (i) Conc. H_2SO_4 at 440K
 - (ii) CH_3COOH
 - (iii) CH_3MgBr
22. What are Amines ? How are they classified ? Give one example of each type.

Or

How is aniline prepared from (i) Nitrobenzene (ii) Benzamide (iii) Phenyl cyanide ?

XIIKDAR21—5002—B

B-2-B<https://www.jkboseonline.com>

(5)

23. What are Carbohydrates ? How are they classified on the basis of hydrolysis products ?

Or

Differentiate between fibrous and globular proteins.

24. Define the following with examples :

- (i) Addition polymerisation
- (ii) Condensation polymerisation

Or

What is Nylon-66 ? Give its preparation and uses.

25. Name the substance which can act both analgesic and antipyretic as well. How does it help heart patients ? What precautions must be taken while taking it ?

Or

Describe the following with suitable examples :

- (i) Artificial sweeteners
- (ii) Antioxidants